

Here are the performance tables from the unclamped simulation, followed by a complete analysis and discussion of the results.

Performance Results for Unclamped PID Controller

Performance at Noise Level $\sigma = 0.0$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.020 ± 0.000	0.036 ± 0.000	0.0043 ± 0.0000
-0.9443	0.026 ± 0.000	0.074 ± 0.000	0.0082 ± 0.0000
0.1559	0.007 ± 0.000	0.012 ± 0.000	0.0020 ± 0.0000
-0.6195	0.023 ± 0.000	0.049 ± 0.000	0.0056 ± 0.0000
-0.8419	0.025 ± 0.000	0.066 ± 0.000	0.0074 ± 0.0000

Performance at Noise Level $\sigma = 0.001$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.020 ± 0.000	0.036 ± 0.000	0.0043 ± 0.0000
-0.9443	0.026 ± 0.000	0.074 ± 0.000	0.0083 ± 0.0000

0.1559	0.007 ± 0.000	0.013 ± 0.000	0.0021 ± 0.0000
-0.6195	0.023 ± 0.000	0.049 ± 0.000	0.0056 ± 0.0000
-0.8419	0.025 ± 0.000	0.066 ± 0.000	0.0075 ± 0.0000

Performance at Noise Level $\sigma = 0.01$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.020 ± 0.000	0.036 ± 0.002	0.0052 ± 0.0001
-0.9443	0.026 ± 0.000	0.074 ± 0.003	0.0091 ± 0.0001
0.1559	0.007 ± 0.000	0.015 ± 0.002	0.0031 ± 0.0002
-0.6195	0.023 ± 0.000	0.050 ± 0.000	0.0065 ± 0.0001
-0.8419	0.025 ± 0.000	0.067 ± 0.001	0.0084 ± 0.0001

Performance at Noise Level $\sigma = 0.1$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.019 ± 0.001	0.069 ± 0.006	0.0166 ± 0.0008
-0.9443	0.027 ± 0.003	0.093 ± 0.009	0.0199 ± 0.0024
0.1559	0.006 ± 0.001	0.057 ± 0.002	0.0154 ± 0.0003

-0.6195	0.021 ± 0.002	0.077 ± 0.009	0.0175 ± 0.0018
-0.8419	0.043 ± 0.025	0.098 ± 0.011	0.0197 ± 0.0006

Analysis and Discussion

This analysis covers the **unclamped** PID controller ($K_p=2$, $K_i=0$, $K_d=0.3$). By allowing unlimited torque, the controller's characteristics change dramatically, resulting in a system that is incredibly fast and astoundingly robust to noise, a stark contrast to the clamped version.

Unmatched Speed and Precision (Low Noise: $\sigma = 0.0$ to 0.01)

In noise-free and even moderate-noise environments, the controller's performance is phenomenal and remarkably consistent.

- **Extreme Responsiveness:** The stabilisation times are exceptionally low, measured in hundredths of a second (e.g., 0.026s from an initial angle of nearly -1 radian). This is because the controller can generate an enormous initial torque to counteract the error almost instantaneously.
- **Controlled Overshoot:** Unlike the clamped controller, this version has a measurable but very small overshoot. This indicates a slightly **underdamped** response, which prioritizes speed. The controller intentionally overshoots the target slightly to get there faster, confident that it has the power to correct the overshoot immediately.
- **Astounding Noise Immunity:** Most impressively, the controller's core performance metrics (**stabilisation time and overshoot**) remain almost completely unaffected even at a noise level of $\sigma = 0.01$. Where the clamped controller began to fail and become inconsistent, the unclamped version's performance is unwavering. It has the authority to simply overpower any incorrect control signals induced by the noise.

Graceful Degradation (High Noise: $\sigma = 0.1$)

This is where the true strength of the unclamped controller is revealed. Where the clamped version failed completely, this controller continues to function, merely showing a graceful

degradation in its precision.

- **Stability is Never Lost:** The controller **successfully stabilizes** in all test cases, even under high noise. This is the single biggest difference from the clamped controller.
- **Precision vs. Noise:** The steady-state error increases by another order of magnitude, as expected. The controller is now constantly fighting the high-frequency noise, resulting in a visible "jitter" around the zero point, but it never loses control. Variability also starts to appear, especially in the stabilisation time for larger initial angles (0.043 ± 0.025 s), but the system remains fundamentally stable.

Conclusion: Clamped vs. Unclamped

This comparative analysis showcases a crucial engineering trade-off between theoretical performance and practical constraints.

- **The Unclamped Controller is a "Brute Force" Solution:** It succeeds by having infinite power. It can correct any error and reject any disturbance because it has unlimited torque. This makes it incredibly fast and robust in a simulation, but it is **physically impossible** to build. Any real-world motor would be saturated, overheated, or destroyed by the immense and instantaneous torque demands.
- **The Clamped Controller is a "Real World" System:** Its failure was not necessarily a result of poor PID tuning, but of **insufficient authority**. The 2.0 torque limit was simply not enough to overcome the destabilizing effects of noise.

Ultimately, the unclamped results demonstrate that the PID gains are fundamentally sound. The engineering challenge is not just to tune the controller, but to manage its limitations. The path forward is not to build a motor with infinite torque, but to **improve the clamped controller's intelligence**—primarily by adding a **filter** to the sensor input. A filter would remove the noise *before* it reaches the PID controller, allowing the safe, clamped system to operate effectively without being misled by spurious signals.